

IT Solutions for a Mobile Clinical Workforce

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The Paper in our Work

- *10- the avg. number of people who touched each paper nursing form at VNAHHS*
- *8- the number of FTEs who shuffled paper forms as their sole job*
- *$60,000 -dollars spent last year on VNA paper forms*
- *9,000 – number of unused and outdated forms in storage*
- *18- avg. hours of paperwork per homecare case*
- *6.50- cost of paper in each medical record*
- *XXXX- records in storage or circulation*
- *XXXXX- avg. cost of storing one record the mandated time period*



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Project Objectives

- Improve Documentation
  - Speed & Accuracy
- Better Quality of Care
  - Reference Databases - Information at the touch of the Screen
  - Interdisciplinary communication
- Improve Analysis/Decision Support
  - Cut the paper chase
  - Improve Productivity
  - Devise an attractive pay structure



Minimize your paperwork.

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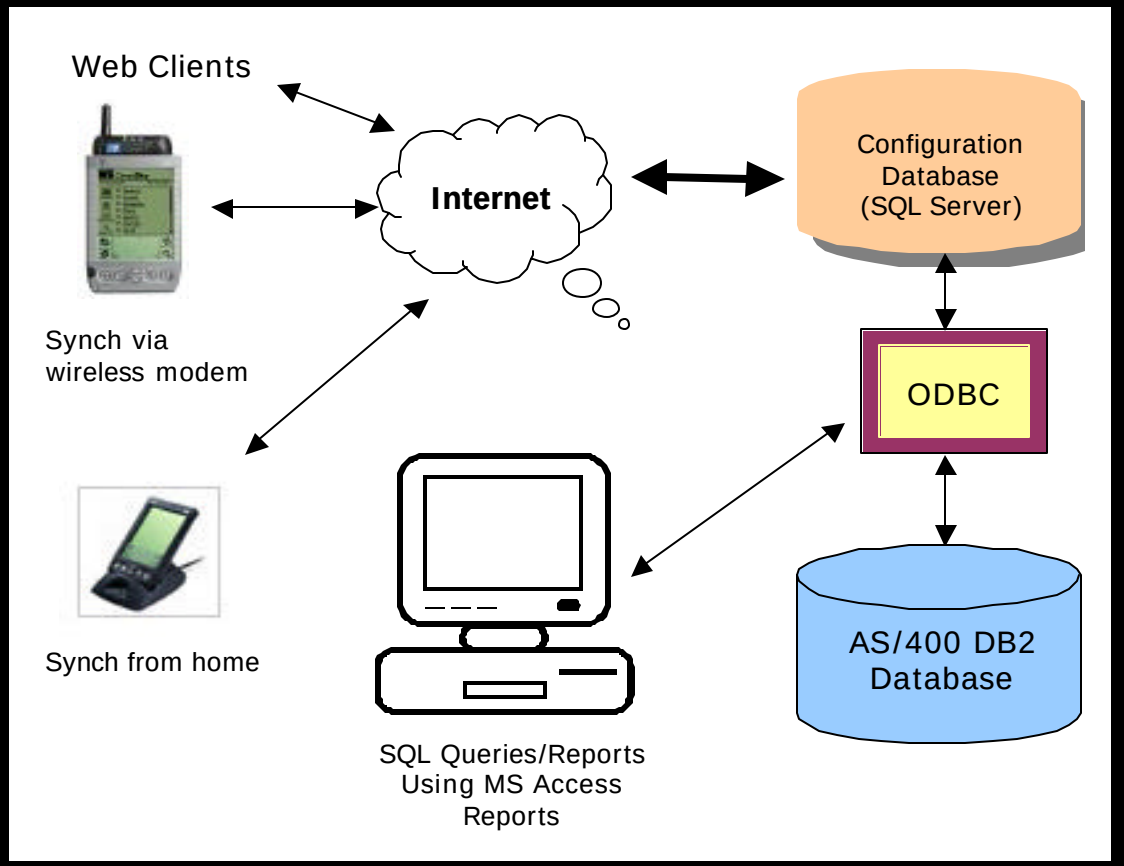
Technology Solution Wish List

- Truly Mobile - Handheld based
- Rapid Database Development & Deployment
- Based on Proven Technology - Standards
- Highly Secure
- Internet Enabled - Access from anywhere at any time
- Synchronization over TCP/IP - not a modem pool
- Affordable
- Easily Manageable
- Highly Scalable
- Minimize Keyboarding/ Maximize Tapping
- Map to external data sources
- Provide for easy data manipulation
- Minimize agency obligation to ‘equip the world’

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How It Happens

- Pendragon Internet Forms
- VNAHHS FreeForms™
- Handheld Device
- Pendragon SyncServer
- FreeForms™ Manager



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Basic Equipment-Mobile Component



- 8 MB Handheld with Cradle
- Portable keyboard
- Nursing Rx and Interact
- VNAHHS reference information
- Wired or wireless modem attachments, if necessary
- Internet capable PC for synchronization of clinical data
- User name and password permission to utilize system

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Web-based Component

- Forms Development and administration
- Secure Access to synchronized data via a web browser



Desktop Component

- VNAHHS FreeForms Manager ™
  - Access to Patient Digital Chart
  - QA functions to check integrity of synchronized data
  - On-demand Reports
  - Electronic submission of documents to third party entities
  - Time/Attendance

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The First Year

Forms Development

Automated paper forms.

1/01

Usability testing to test for skip logic, terminology, etc..

Sophisticated pick lists to improve documentation quality

Developed feedback loops from server to pda to populate forms with demographic data

Implemented electronic chart format to all clinicians

Mapped supplies ordering and scheduling downloads.

Initiated first 'outside the agency' connection via the system.

1/02



Install Pendragon Forms 3.1

Tested Synchronization software

Purchased additional licenses. Go live with PC based forms software.

Installation of Pendragon Internet Forms software

Testing:
--Synchronization
--Security
--Mapping
--Reporting

Go-live on Internet Forms with supplies ordering and non-patient data collection forms.

Infrastructure Set up

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Lessons Learned



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#1 Understand Capabilities/Constraints

- The Medium - 4.8” x 3.0” x 0.7”
- The Users - Nurses by profession not computer geeks
- The Software

#2 Consider Important Trade-offs

Speed
Standardization
Validations



Accuracy
Individuality
Flexibility

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#3 Give Priority to Data Quality

Data quality starts with the clinician's stylus and never gets any better!



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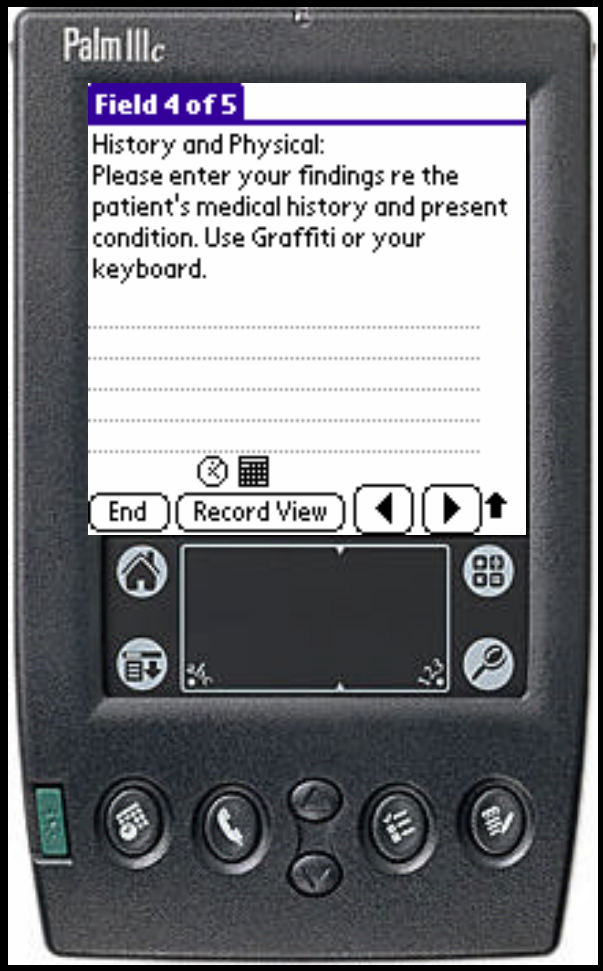
#4 Turn Data Into Information

- Take advantage of your digital assets for decision support
- Use information to derive knowledge for competitive advantage
- Provide managers with tools to get to the data
- Develop new QA measures

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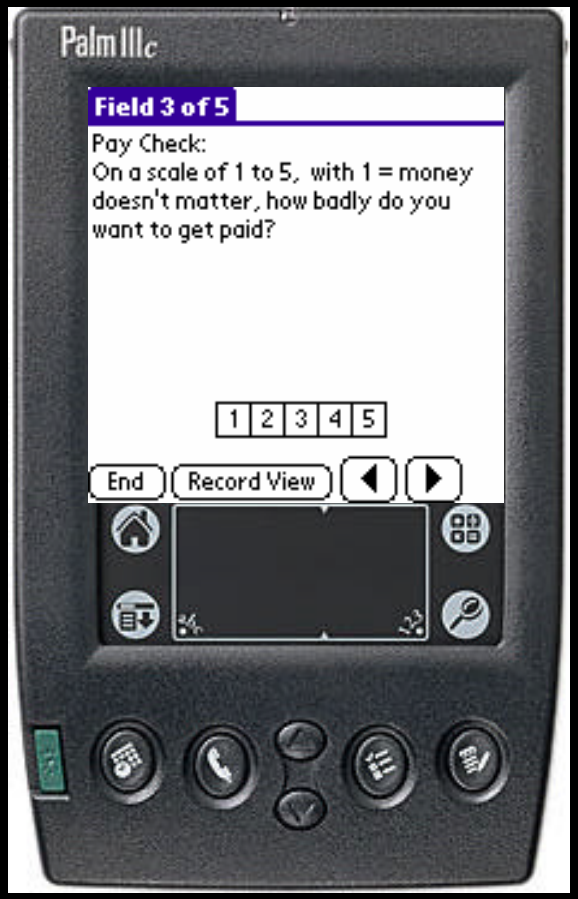
#5 Re-engineer Business Processes

- Do not just turn paper documents into electronic forms
- Re-engineer before designing forms



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#6 Do Not Automate Everything



- Some things DO NOT work best on a handheld!
- Consider the customer's needs
- Don't try to solve management problems with software
- Never ask a question for which you already have an answer

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#7 Design Forms for Clinicians by Clinicians



- Encourage in-house development
- Create validation points that display at the time of entry
- Message box reminders
- Take advantage of reference databases
- Know when to draw the line....

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#8 Make Training & Tech Support a Critical Success Factor

You are how you act

You are how you sound

Mix up the gene pool

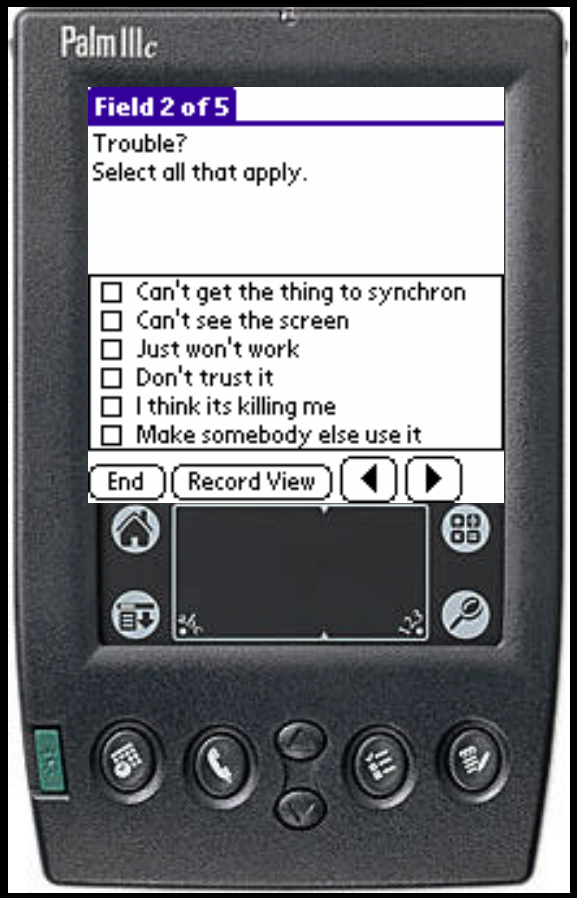


#9 Protect the Integrity of the Database

- Set up constraints on user permissions
- Restrict modification of data after submission/synchronization
- Implement new policies and procedures
- Assure confidentiality - Users must trust the quality & integrity of the data

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#10 Be Patient



- Use a phased rollout approach
- Resist the temptation of wholesale conversion



Evaluation

- Do the widgets work?
- Recruiting/retention statistics
- Research
  - Speed
  - Accuracy
  - Satisfaction
- Office efficiencies

What's Next?

- Tele-monitoring interface
- Improve reach outside the organization
- Automated Scheduling
- OASIS on Palm OS® devices



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